

What the Data Says when Country Listens: Patterns of Priority and Engagement Among Indigenous Australian Voters

Part 0: Introduction

0.1. Background

In October 2023, Australians voted on whether to enshrine an Indigenous "Voice to Parliament" advisory body in the Constitution. It failed nationally (60% No), but Indigenous communities voted strongly Yes. The 2025 election was the first federal election after this loss. The Aboriginal communities experienced grief over this outcome, stating that most Australians does not understand the struggle and concern of The Indigenous Communities.

We offer to use Indigenous voters 2025 data to understand the indigenous communities better. There are three layers we use to bridge this understanding:

- 1. Layer 1 — What matters?** This layer will discuss which issue priorities and concerns travel together in this community. We expect the result to tell policy makers that addressing one issue without its companion issues will likely feel inadequate to the people affected.
- 2. Layer 2 — What separates engaged from disengaged voices?** This layer will inspect what factors distinguish those who feel the system works for them from those who don't among Indigenous voters. It should inform tells policymakers where the listening gap is widest.

In doing this project, we recognise that data cannot fully capture the struggles and concerns of the Indigenous community. Lived experience and voices from the community remain indispensable to Australian policymaking, and the structural channel that the 2023 referendum proposed for these voices remains needed. This project does not propose itself as an alternative to that listening. Its purpose is smaller: to offer some relief from the grief that followed the referendum's defeat by showing that what the community has been communicating is still legible, still coherent, and still available to be acted upon.

0.2. Data Assessment

A. Data Source

Data that is used in this essay can be found in:

<https://dataverse.ada.edu.au/dataset.xhtml?persistentId=doi:10.26193/A9VEXA>

B. Data Summary

The dataset is well-curated and high-quality. Completeness is excellent: most key analytical variables have zero missing responses. A more important data-quality consideration for this project is response symmetry. Many analytical variables lean strongly toward one response option even when no single bucket reaches the strict 50% criterion used in the table below — a substantial number of items show 40–45% concentration in one category, particularly the Indigenous-issue importance items and

several social-attitude items. The Q40 series shows the strongest case of this, this aspect is one of key consideration in choosing analytics method.

For the purposes of this project, it is useful to divide questions into four category and evaluate quality of each category:

Bucket	Description	Missing	Highly Biased Variables (>50% Response in 1 bucket)
Demographics	Confidentialised codes for gender, age band, education, and state of residence	Gender: 3% Age:12% Education: 19% State: 2%	None
Political choices	Q1, Q3, Q13, Q15, Q16, Q21, Q34_1, Q34_2 Vote choices, engagement, left-right placement, trust, and civic attitudes	Q1:0% Q3: 0% Q13: 0% Q15: 9.9% Q16: 9.9% Q21: 9.9% Q34_1: 0% Q34_2: 0%	Q15: 418 Vote on “Labor Party” Q15: 411 Vote on “Labor Party” Q21: 467 Vote on “Labor Party”
Issue priorities & policy	Q35, Q36, Q38, Q40-Q46 Top-issue selections and agree/disagree positions on economic, social, and Indigenous-policy items	Q35: 0% Q36: 0% Q38: 0% Q40: 0% Q41: 0% Q42: 0% Q43: 0% Q44: 0% Q45: 0% Q46: 0%	Q35: 471 votes on “cost of living” Q40_1: 621 Vote on very important Q40_2: 571 Vote on very important Q40_3: 606 Vote on very important Q40_4: 473 Vote on very important Q40_5: 579 Vote on very important
Party-issue alignment	Q37_1 to Q37_10 — which party's policies are closest to the respondents on each major issue	Q_37:0%	None

Part 1: Layer 1 – What Matters?

1.1. Why Issues Travel Together

The first analytical layer asks how respondents' concerns cluster together. Rittel and Webber (1973) argued that argue that social processes act as links tying "open systems into large and interconnected networks. Following this logic, the motivation for layer 1 is a foundational claim about social problems: they are not the sum of their component parts. Acting on any one in isolation typically fails not because the action was poorly

designed, but because the conditions that produced the original problem remain intact through the other channels.

This concern has particular weight after the 2023 Voice referendum, where the country's tendency to address Indigenous concerns one item at a time risks repeating an approach the community has already watched fail.

By analysing issues as facet of systemic concern instead of individually, we find pattern that open the opportunity for meaningful discussion, such as voter alienation, systemic vs individual concern, and ideological coherence

1.2. Data Analysis

1.2.1. Analysis Method

Model Selection:

Layer 1 is a question on *co-occurrence pattern*. Association rules, more specifically with Apriori algorithm, is uniquely suited to surface which combinations of beliefs co-occur more often than chance, without imposing structure on the data, and its rule output is directly interpretable for policy makers.

Analysis variables: Question on **Issue priorities & policy** Discussed in section 0.1.

Software: R, using 'arules' package to create association rules.

Parameter Tuning

There are two parameter we need to choose for this model: support and confidence. Parameter tuning as done in machine learning approaches does not work in this context, as we do not have objective loss function. However, the big purpose of this project on listening shapes both parameters' choices. The support threshold must be set low enough that minority belief bundles within the community can surface if they exist. The confidence threshold faces a different problem. We must avoid too high threshold that result in near-tautological rules, while keeps it high enough to ensure strong association rules. The thresholds used in this analysis (5% support corresponding to ~40 of 802 respondents, 60% confidence, maximum rule length 5) sit inside this reasoning rather than being defended as universally correct and were applied uniformly across all three subsets to ensure cross-subset comparability.

Analysis Process:

1. Conversing data into 'shopping bag'-like structure for association rules

Each respondent forms one basket; each strongly held view becomes one item. Likert items (Q38, Q44) split into =agree (values 1–2) and =disagree (values 4–5); importance items (Q40) split into =important and =not_important; categorical items (Q35, Q36, Q42) record the chosen option. Neutral midpoints and missing values contribute no item. For example, the shape of the basket would be:

["Q35=cost_of_living", "Q36=health_medicare", "Q38_1=agree"].

No missing data handling was done in this part, as all variable have 0% missing.

2. Mining frequent rules using the Apriori algorithm with two thresholds: each pattern must clear the threshold for support and confidence. Rules longer than

five items were excluded. Duplicate or weaker versions of the same pattern were removed, and what remained was ranked by lift.

3. **Comparing rule structure across three subsets:** Ran the same mining process on three groups: all respondents, Voters who voted yes in the 2023 referendum, and Voters who voted no in the 2023 referendum. From each group, the top 50 rules by lift were extracted along with their support, confidence, and other quality scores. Items appearing on the left-hand side of these rules were counted to see which beliefs characterise each group.

***Note:** in further analysis we will address Voters who voted yes in the 2023 referendum as Voice=Yes and, vice versa, Voters who voted no in the 2023 referendum as Voice=No for simplicity.*

4. **Assessing pattern robustness:** Pattern strength was assessed two ways: (i) comparing lift values against the maximum possible lift for each rule's outcome (since common outcomes like Voice=Yes have a lower ceiling than rare ones); (ii) measuring how much the rules overlap by checking whether they describe the same respondents (Jaccard similarity).
5. **Find the top 50 rules for all group [here](#).**

1.2.2. Analysis & Findings

LHS items	RHS	Support	Confidence	Lift	Count
Q38_5=disagree, Q38_6=agree, Q40_1=important, Q40_5=important	Q38_2=disagree	0.051	0.932	4.73	41
Q38_3=agree, Q38_5=disagree, Q38_6=agree	Q38_2=disagree	0.052	0.913	4.63	42
Q38_5=disagree, Q38_6=agree, Q40_5=important	Q38_2=disagree	0.052	0.913	4.63	42
Q38_5=disagree, Q38_6=agree, Q40_1=important	Q38_2=disagree	0.051	0.911	4.62	41
Q38_5=disagree, Q40_1=important, Q44_8=agree	Q38_2=disagree	0.056	0.900	4.57	45

TOP 5 Association Rules for all respondent, sorted by Lift

A. The People who voted No in 2023 Referendum Have no Shared Believe

No association rule for the Voice=No subset cleared the 5% support threshold (Section 2.2.2, item 5). Given the size of this group (~158 respondents, ~25% of valid responses), the absence is informative rather than incidental: no belief bundle is shared by a substantial fraction of No voters, suggesting this group is internally too fragmented to characterise as a unified constituency.

This shifts the strategic calculation. Standard political response to an opposing constituency assumes shared preferences that can be addressed or partially accommodated. The data suggests no such unified Voice=No constituency exists to address. A government oriented toward the substantive agenda already supported broadly across the community, including by most No voters on items other than the constitutional question, therefore faces limited risk of systemic voter alienation from the

No-vote side, **opening political space for action on the Yes coalition's continuing priorities** wider than a conventional reading of the referendum result might suggest.

B. For Indigenous Community, Structural framing predicts coalition support; individual framing does not

The strongest patterns in the full sample point to one underlying signal: strong pro-union sentiment. Forty-eight of the top fifty rules predict Q38_2=disagree (rejection of the view that "trade unions have too much power"), and all of them include Q38_5=disagree (rejection of the related claim that "there should be stricter laws to regulate trade unions") on the left-hand side. The single-item rule {Q38_5=disagree} => {Q38_2=disagree} already achieves confidence 0.83 and lift 4.21, covering 68 respondents (8.5% of the sample). Subsequent rules in the top fifty add further items as refinements on this anchor pair, narrowing the rule down to smaller and more ideologically consistent subgroups. The tightest bundles cover roughly 5% of the sample at confidence above 0.90 and lift approaching 4.7.

In the survey, Q40 specifically asked for the concern of Indigenous community, specifically, and how many times they showed up in each group top rules:

Item	Description	Frequency in all rules	Frequency in voiceYes rules
Q40_1	Land rights	16	1
Q40_2	Constitutional recognition	2	13
Q40_3	Self-determination	12	1
Q40_4	Welfare dependence	1	1
Q40_5	Social disadvantage	7	2

The notable exception is welfare dependence. Q40_4 appears in only one of the 48 rules. Constitutional recognition (Q40_2) also appears only once, but for a different reason — Q40_2 is a Voice=Yes-specific item rather than a general progressive marker.

Progressive respondents engage with Indigenous issues framed in structural terms but disengage from those framed individually, even when the underlying policy is identical. The implication for government communication is direct: framing matters as much as substance. An **initiative described in structural terms** (addressing dispossession, exclusion, or systemic disadvantage) aligns with how this coalition thinks about Indigenous policy. The same initiative framed around welfare dependence will not.

C. Indigenous-policy salience is not what mobilises the Voice=Yes coalition, and policymakers should not assume it is

Meriam webster dictionary define social-liberal policy as a political philosophy stressing the importance of the government's role in promoting the public good, relieving social and economic inequities, and protecting and supporting marginalized people. In our data, the preference for social liberal policy shown as:

Item	Question	Progressive position
Q43	Same-sex marriage	Favour
Q44_1	Death penalty for murder	Disagree (oppose)
Q44_5	Turn back asylum boats	Disagree (oppose)

Item	Question	Progressive position
Q44_6	Voluntary assisted dying	Agree (support)
Q44_7	Increase opportunities for women in business	Agree (support)
Q44_8	Allow pill testing at music festivals	Agree (support)
Q42	Abortion	"Available readily"
Q45	Immigration numbers	Increase
Q46	Climate change as a threat	Serious

Of 50 top Voice=Yes rules:

- 50 contain at least one social-liberal item
- 18 contain at least one Indigenous-issue item
- **0 contain only Indigenous-issue items (with no social-liberal items)**
- 18 contain both

The implication for policymakers is that an "Indigenous-issue salience" framing is not the primary mobilisation channel for this coalition. Political communication research suggests that for ideologically consistent voters, single-issue appeals are less effective than appeals that engage the broader ideological frame. **Talking to the community as "you who care about Indigenous people" works less well compared to "you who hold this set of progressive commitments, of which Indigenous policy is one."** This sounds like a small framing distinction, but it changes how a government communicates and which coalition it speaks to. The Yes coalition cares about Indigenous issues, but they care about Indigenous issues because of the broader worldview, not the other way around. Indigenous-policy progress benefits from being communicated within that broader worldview, not extracted from it.

Part 2: Layer 2 - What separates engaged from disengaged voices

2.1. Political engagement among Indigenous Australian

Layer 1 mapped what this community believes. Layer 2 turns to how it experiences participation in Australian democracy: whether people feel the political system is working for them, and where the gap between formal enfranchisement and felt participation is widest.

The 2023 Voice referendum was, in part, an offer of structural participation that the country declined. That refusal carries weight in a community where engagement already varies substantially. Whether the refusal is reflected in democratic satisfaction in 2025 is exactly the question Layer 2 asks. If satisfaction tracks engagement and partisan factors more strongly than Indigenous-specific concerns, the listening gap is not uniquely Indigenous but structural to the political process itself. Layer 2 identifies where that gap is most clearly located.

Data Analysis

2.1.1. Data & Method

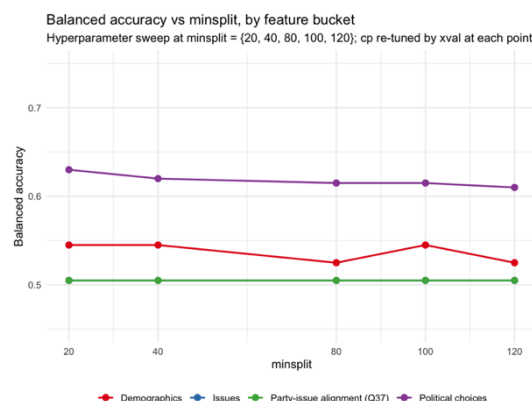
Model Selection

The skewed distributions documented in Section 0.1 make decision trees a natural fit for this analysis. Tree-based methods make no parametric assumptions about predictor distribution. Second consideration is interpretability. The audience for this report's findings is not data-mining specialists but policymakers, advocates, and community members. Decision tree is one of the easiest models to be interpreted by non-technical audience.

Parameter Tuning

Decision tree has several interchangeable hyperparameters: maximum depth, minimal leaf, minimal split. In this experiment, we pick minimal split as training hyperparameter for the reason of interpretability as this maps directly onto a substantive question: "how small a respondent subgroup are we willing to characterise as a distinct group?". The best parameter is chosen through experimentation, using grid-search of [20,40,80,100,120] with 10-fold validation, and accuracy (accuracy=(TP+TN)/Total observation) as objective function.

Grid search result on 4 trees:



Note: Metrics on Issues & Party alignment issues are overlap, hence the plot only shows 3 lines.

Software: R, using rpart for decision-tree fitting and caret for evaluation metrics.

Analysis Method:

1. **Missing data** was imputed with codes (e.g. 97, 98, 99, 9999, -1, -8, -9) across Likert-scale and party-alignment variables or NA for demographics.
2. **Building the target.** Democratic satisfaction was derived from Q27 ("are you satisfied with the way democracy works in Australia?"). Respondents who answered, "very satisfied" or "fairly satisfied" (values 1, 2) were labelled **satisfied** (n ≈ 333, 42% of sample); respondents who answered "not very satisfied" or "not at all satisfied" (values 3, 4) were labelled **dissatisfied** (n ≈ 469,

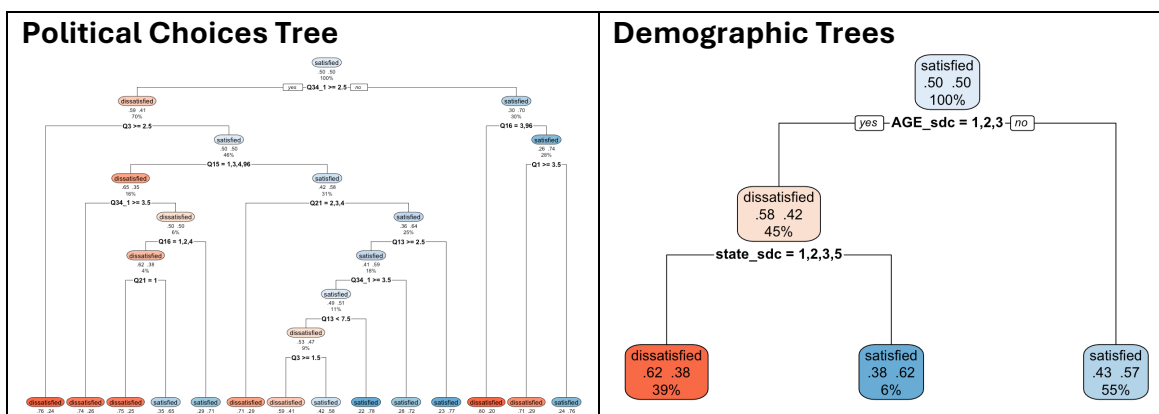
58%). Q27 was chosen as it was a reasonable proxy to measure engagement and balance between two classes are quite good.

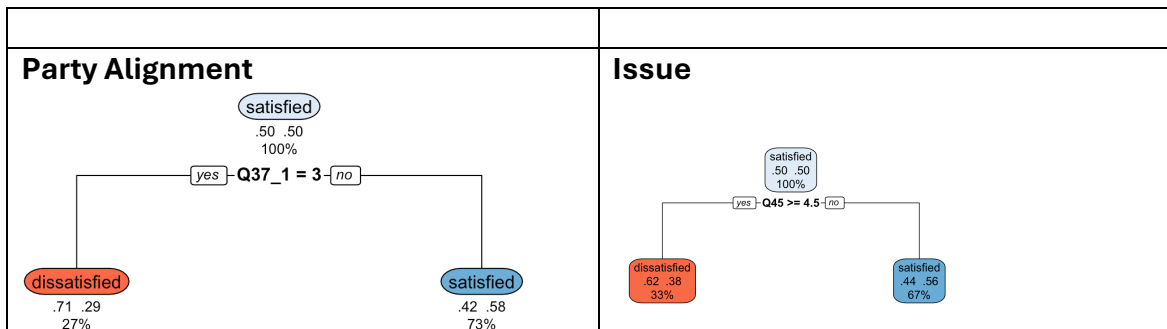
3. **Defining four feature buckets.** To identify where the listening gap between satisfied and dissatisfied Indigenous voters is widest, the available predictors were split into four conceptually distinct buckets, as explained in section 0.1.
4. **Fitting four trees with identical conditions.** A single train/test split (75/25) was generated once and reused across all four trees so that any difference in predictive performance is attributable to the predictor set rather than to differences in sample composition. Each tree was fitted using rpart, ten-fold cross-validation for complexity tuning. All trees used the same algorithmic parameters.
5. **Comparing predictive performance.** Each tree was evaluated on the held-out test set using accuracy, sensitivity (correctly identifying satisfied respondents), specificity (correctly identifying dissatisfied respondents),
6. **Interpreting the winner through variable importance.** For the winning tree, the variable importance ranking was extracted to identify which specific items inside the bucket drove the prediction. Pre-pruning importance was also recorded so that variables tested but removed during pruning are not silently mistaken for variables the algorithm never considered.

2.1.2. Result & Analysis

Across four decision trees fitted on identical data with identical splits, political-choices factors are the strongest predictor of democratic satisfaction in this Indigenous sample, achieving accuracy of 0.63 against 0.55 (demographics), 0.51 (party-issue alignment), and 0.51 (issues). No other feature bucket comes close on overall predictive power.

Feature bucket	Features	Best Min-split	Accuracy	Sensitivity (<i>satisfied</i>)	Specificity (<i>dissatisfied</i>)	Top Predictor
Political choices	8	20	0.63	0.58	0.67	Q3
Demographics	4	100	0.55	0.63	0.49	Age
Party-issue alignment (Q37)	10	120	0.51	0.72	0.35	Q37_1
Issues	27	120	0.51	0.72	0.35	Q45





Three observations:

First, engagement and vote choice dominate. The top four predictors (campaign interest, House vote, political interest, Senate vote) are all about how connected respondents are to the political process and which side of it they sit on.

Second, the Indigenous-specific items collapse to near-zero in the political choice tree. Q21 (Voice vote) and Q34_2 (Indigenous civic attitudes) appear at the bottom, effectively unused by the tree once engagement and vote are accounted for. This is the single most important sub-finding inside the political choices result.

Third, the issues tree's poor performance is itself informative. Even with 27 issue-related features available, the issues tree barely outperforms random guessing (accuracy 0.51). Substantive policy positions do not predict democratic satisfaction in this sample.

Implication: The listening gap is in the political process, not in policy substance. André, Audrey & Depauw (2017) support this idea that satisfaction in government does not come from quality of policy alone. Engagement is the dominant predictor of democratic satisfaction: Q3 (campaign interest) outranked Q15 (House vote) by nearly 2-to-1, and even Indigenous-issue importance, land rights, self-determination, social disadvantage, does not predict whether respondents feel democracy is working for them. **A government that delivers excellent Indigenous policy while failing to engage Indigenous voters in the political process will not see democratic satisfaction improve.** The pathway runs through engagement: voter outreach, campaign accessibility, and Indigenous-led civic education are the levers that move this needle, and no other alternative suffices.

Part 3: Conclusion

This project began with a question about grief. The premise was modest: that the data already collected from Indigenous voters in 2025 contains a great deal of communication, and that surfacing it carefully might contribute something toward the listening the referendum was meant to formalise.

Three findings emerged. **First, the community is not divided on the substantive Indigenous-policy agenda, only on a particular constitutional mechanism.** Concern with land rights, self-determination, and social disadvantage is held broadly across the community regardless of Voice vote; the agenda survives the referendum's defeat. **Second, Concerns travel together, and framing determines which coalitions engage with them:** systemically framed Indigenous issues bundle with the progressive worldview, individually framed concerns like welfare dependence do not, and the same

policy presented in either framing will reach different audiences. **Third, the listening gap is in the political process itself:** democratic satisfaction is best explained by engagement and partisan factors, not by issue positions or Indigenous-specific civic concerns, meaning policy delivery alone will not repair it, the pathway runs through connection to the political process.

The grief of 2023 was not the grief of having been wrong about what mattered. It was the grief of having been heard and refused. This project cannot undo that refusal, but it shows that what the community communicated remains legible, coherent, and available to be acted upon.

The data shows a community that has been speaking clearly. The work that remains is the listening.

Part 4: Further Work

Surveying the disengaged. The finding that engagement predicts democratic satisfaction more strongly than any other factor raises an immediate question about who is missing from this dataset. The most disengaged Indigenous voters — those least likely to complete a political survey — are also those for whom the listening gap is likely widest. Community-led research methods designed to reach the unenrolled and the non-respondent would extend the engagement finding toward the voices this analysis could not reach.

Pre- and post-referendum comparison. The finding that the substantive Indigenous-policy agenda survives the referendum's defeat is based on 2025 data alone. Pairing this dataset with pre-referendum surveys from the Australian Election Study would test whether the bundling patterns and policy priorities identified here were stable across the referendum or whether the campaign and its loss reshaped them. Specifically: did the structural-vs-individual framing asymmetry exist before the Voice campaign, or did the campaign itself sharpen it?

Qualitative work alongside the patterns. The finding that framing determines which coalitions engage (structural terms mobilise the progressive coalition, individual terms do not) identifies a pattern but cannot explain it. Interview-based research with respondents who hold the structural framing would test whether the mechanism is ideological (a principled rejection of individual-blame narratives) or experiential (a lived recognition of structural disadvantage). That distinction matters for how the finding is applied in practice.

The findings of this project are partial by construction. They describe what one dataset can show about one community at one moment. The work of listening that the project began is, by its nature, ongoing.

References

Han, J., Pei, J., & Tong, H. (2023). Data mining : concepts and techniques (Fourth edition / Jiawei Han, Jian Pei, Hanghang Tong.). Morgan Kaufmann.

Rittel, H.W.J., Webber, M.M. Dilemmas in a general theory of planning. Policy Sci 4, 155–169 (1973). <https://doi.org/10.1007/BF01405730>

Mason, Lilliana. (2018). Ideologues without Issues: The Polarizing Consequences of Ideological Identities. Public Opinion Quarterly. 82. 866-887. 10.1093/poq/nfy005.

André, Audrey & Depauw, Sam. (2017). The Quality of Representation and Satisfaction with Democracy: The Consequences of Citizen-Elite Policy and Process Congruence. Political Behavior. 39. 10.1007/s11109-016-9360-x.